

NAME: _____

PERIOD: _____

Same Responses, Different Questions

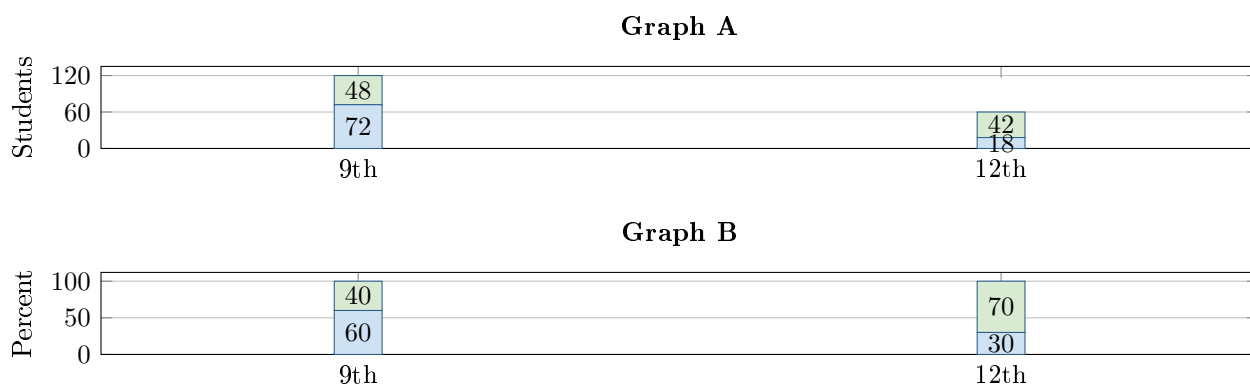
AP Statistics · Unit 2 · Topic 2.1

[STAR] THE PROBLEM

Suppose a student council asks 120 ninth graders and 60 twelfth graders whether they prefer Field Day or a Talent Show for the spring celebration. The same responses are shown in two graphs.

Andre: *Graph B hides how many people there are because both bars reach 100%.*

Leila: *Graph A hides the relationship because the grade totals are different.*



Lower: Field Day Upper: Talent Show

FIRST TAKE — one sentence before you work the parts

For the question *Is celebration preference related to grade level?*, which graph would you use first? Name one visible feature.

[BOOK] MATCH THE DISPLAY TO THE QUESTION (keep on the page)

A count bar graph shows numbers of students. A segmented bar graph stacks each group's percentages to 100%. To compare group preferences, divide each count by its group total and compare the within-group percentages. Different group distributions suggest association. Percentages alone do not show how many people were surveyed.

Work (a)–(c).

DOK 1 (a) From Graph A, state the total surveyed in each grade.

DOK 2 (b) Find the percent who prefer Field Day in each grade. Show the fractions and compare the results.

DOK 3 (c) Which graph better shows whether grade and preference are related? Use both Field Day percentages and explain how the bar heights help. Then explain what Andre is right about and one question Graph A answers more directly. Write 3–4 sentences.

Frame: Graph _____ better answers the relationship question because equal-height bars let me compare _____.

Frame: Graph A preserves _____, while Graph B cannot tell me _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.